

# Recording a Route Without Satellites

Speed from Vibration and Direction from Turns on a Smartphone, Measured Against GPS

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## Abstract

When GPS is lost — in a tunnel, an underground car park or an aircraft — a phone can still draw the route it travels. This paper describes a system that reads *speed* from how the vehicle shakes and *direction* from the phone’s gyroscope, accelerometer and magnetometer, using GPS for nothing but the first point, and measures it against GPS on a motorcycle, in a car, on foot and on a plane. Every recording was replayed through the newest version from its raw sensor data, with GPS used only as the answer key: 103 motorcycle and car journeys (30 hours, 548 km that GPS could check), 103 straight stretches and 7 longer walks on foot, and one flight. Direction was within  $30^\circ$  of GPS 76% of the time on a motorcycle, 87% in a car, 93% on foot and 99% on the plane, and the median route was turned  $9^\circ$  on a motorcycle and  $13^\circ$  in a car. Against GPS, distance came out  $-12.4\%$  on a motorcycle,  $-2.6\%$  in a car and  $+2.6\%$  on foot. In the air, a network trained on 302 airline flights from NASA’s public flight recorder data, reading only the time since the takeoff roll and the phone’s tilt, raised the flight’s distance from 61% to 103%, and a store of examples from the same data to 99%; drawn with no GPS after the first point, the flight ended 16.9 km from where GPS did. A phone with no examples of its own starts with a small built-in network, which on recordings it had never seen was a little more accurate than a full store; both built-in networks are drawn layer by layer. We also report what each part of the direction method adds, where the system fails, and the approaches that did not work.

**Keywords:** dead reckoning; smartphone sensors; GNSS-denied navigation; vehicle speed estimation; heading estimation; magnetometer

## 1 Introduction

A phone that loses GPS still knows a great deal about how it moves. Its accelerometer feels every push and every bump, its gyroscope measures every turn, its magnetometer senses north, and its barometer feels height. In principle that is enough to keep drawing a route through a tunnel, an underground car park or a flight. In practice two things make it hard.

The first is *speed*. Adding up acceleration, the textbook answer, does not work on a phone: its small sensor errors grow into kilometres within minutes [1, 2]. The second is *direction*. The gyroscope measures turns but not north; the magnetometer measures north but in a vehicle it is disturbed by steel, and in any case it gives the direction the *phone* points, not the direction of travel, which differ by however the phone happens to sit in a pocket or a holder [3].

This paper describes a system built around those two problems, running as a mode of an ordinary iPhone app, and measures how well it works. Its contributions are:

- **A complete route recorder that uses GPS for the first point only.** Speed comes from a fingerprint of the vehicle’s vibration, looked up in a store of the phone’s own earlier trips or, on a new phone, answered by a small built-in network. Direction comes from the vehicle’s turns, which reveal how the phone sits, on top of a heading whose slow drift is

measured at stops and whose starting error is corrected by the magnetometer whenever the magnetic field proves to be Earth’s, by holding still in the world while the vehicle turns (Section 3).

- **An evaluation against GPS that reflects the newest version, one kind of travel at a time.** Every recording was replayed through the current method from its raw sensor data (Section 3.5), and each kind of travel has its own section and figures: the motorcycle, walking, the car and the plane (Sections 4.1–4.4). Appendix A lists every journey.
- **How the built-in networks work.** The two small networks that answer before a phone has learned anything — one for speed on the ground, one for the air — are drawn layer by layer, with how they were trained and how well they do (Section 5).
- **An account of what matters.** Every kind of travel side by side, what each part of the direction method adds (Section 6), where the system goes wrong, and the approaches that did not work (Section 7).

## 2 Related work

**Inertial navigation.** Integrating acceleration twice gives position, but the error grows quickly — with the square of time for an accelerometer bias and faster still for a gyroscope bias [1]; smartphone accelerometers and gyroscopes carry biases and noise large enough to make unaided integration useless within minutes [2]. Practical systems therefore constrain the problem.

**Pedestrian dead reckoning.** For walking, the usual constraint is the step: count steps, estimate their length and their direction [4]. A sensor on the foot can reset its velocity at every stance [5]; a phone in a pocket cannot, and its heading depends on how it is held, which several methods try to resolve [3]. Our walking mode follows the step-counting tradition; its direction method is our own.

**Orientation.** Orientation filters fuse gyroscope, accelerometer and magnetometer readings [6], and on an iPhone Apple’s Core Motion provides one. Such filters assume gravity is the only steady force, so a vehicle’s sustained acceleration — a takeoff roll — is mistaken for a tilt [7]. We found two more practical problems with the built-in heading, in a pocket and in a car (Section 3.2).

**Vehicle speed from vibration.** Road vibration changes with speed, and learned models can read speed from accelerometer data alone [8]. Our speed model is of that kind, but it learns continuously from the phone’s own trips and must work for a phone in a trouser pocket on a motorcycle as well as in a car.

## 3 Method

A route is a series of short steps, one per second. Each step needs a length and a direction; with GPS both come from satellites, here both come from the phone (Figure 1).

### 3.1 Speed

**A fingerprint of the vibration.** Every second the phone takes the last five seconds of up-and-down acceleration, measured 50 times a second, and reduces it to eleven numbers: the



**Figure 1.** The system. Speed comes from the accelerometer’s vibration, or in the air from integrating the takeoff and then from a network trained on airline flights; direction combines the angle of the phone, learned from the gyroscope’s turns and the push felt in them, with a heading reference. GPS gives the first point and nothing else.

energy of the shaking in nine frequency ranges and two measures of its overall strength. A vehicle shakes differently at different speeds, and the fingerprint captures that.

**A store of the phone’s own trips.** Whenever GPS is available the phone stores fingerprints with the speed GPS measured at that moment, keeping up to 4,000. Without GPS it looks up the 12 stored fingerprints closest to the current one and averages their speeds, nearer ones counting more:

$$\hat{v} = \frac{\sum_{j=1}^{12} v_j / d_j^2}{\sum_{j=1}^{12} 1 / d_j^2}, \quad (1)$$

where  $v_j$  is a stored speed and  $d_j$  the distance between fingerprints. If even the closest stored fingerprint is far away, or the closest ones disagree among themselves, the store gives no answer rather than a guess, and the starting network below answers in its place: the speed never waits for GPS and never stalls while the vehicle moves. Averaging pulls every answer toward the middle, so a correction learned from the stored examples stretches the answers back out.

**A starting network.** A new phone has no examples, so until it holds 3,000 it uses a small network (11 inputs, two layers of 32, one output) trained on recorded trips and shipped with the app. It needs nothing from the current trip, and it declines a fingerprint unlike anything it was trained on.

**Stops and hands.** A stopped vehicle is recognised from the absence of shaking, not from the speed model, which would read an idling engine as a crawl. A hand on the phone shakes it, and shaking reads as speed, so while the phone is being handled the speed may fall but not rise, and a phone held in the hand keeps the last speed measured before it was picked up.

**In the air.** A smooth cabin reads as standing still. In the air the phone instead measures the takeoff: from the last moment it was still, it follows its own turning with the gyroscope and adds up the push the accelerometer feels for up to two minutes. Core Motion’s own orientation cannot be used for this, because it takes a takeoff’s steady push for a tilt [7]. Integration beyond two minutes gathers too much error, so the phone used to hold the speed it had reached, which misses the speed-up in the climb.

**After the takeoff: two flight engines.** What a phone in a cabin can still sense is how long ago the takeoff roll began and how it is tilted: an aircraft climbs nose-up and levels off as it reaches cruise. The phone measures its tilt once a second against the way gravity pointed at the still moment before the roll, and forms four inputs: minutes since the roll began, the average tilt over the last minute and the last five minutes, and the tilt added up since the roll began. Two engines turn these into a speed, in the same pair as on the ground. The *flight network* has four inputs, two layers of 32 and one output. The *flight store* holds 4,000 examples and answers like the phone's store, averaging the 12 nearest with weights  $1/d^2$ . Both were trained offline on NASA's public flight recorder data (Section 3.4), with the recorded ground speed as the answer. They use no GPS and nothing from the current trip, and inputs beyond the range of the training flights are held at its edge. The network is used, the store stands in if the network is missing, and the air examples from the phone's own earlier flights take precedence when there are any. None of this is used on the ground.

## 3.2 Direction

**A heading reference.** The phone's built-in heading follows a direction inside the phone that depends on how it is held, and when that direction points nearly straight down — a phone head-down at an angle in a trouser pocket — it scattered by about  $49^\circ$  a minute although the phone did not move. The system therefore computes its own heading from the phone's orientation, using whichever edge of the phone lies closest to level; in the same position that heading scattered by about  $1^\circ$  a minute.

**Drift, measured at stops.** In a car the heading also drifts slowly with the gyroscope's own bias, because the steel around it keeps the magnetometer from holding north: on one drive it turned  $42^\circ$  in 25 minutes with the phone fixed in place. A stopped car does not turn, so any turning the heading shows while the car is stopped and the phone still is drift. The phone measures it as a rate, at most  $3^\circ$  a minute, and takes the accumulated drift off every heading. It counts a still second only while the magnetic field reads like Earth's here, 30 to  $60\ \mu\text{T}$ : Core Motion's heading leans on the magnetometer, and in a disturbed field it is pulled toward the disturbance — which in a car is not the same standing as driving, so a drift measured at stops would be carried wrongly into the drive.

**The magnetometer, when the field can be trusted.** Core Motion also sets north once, in the first seconds of a recording, and then follows the gyroscope: on the recordings with the magnetometer logged its north moved by up to  $160^\circ$  in the first second and afterwards seldom more than  $5^\circ$  in a minute beyond what the gyroscope measured. Set inside a vehicle, that north is often wrong. On one drive it began  $20$ – $26^\circ$  off while the magnetometer, read directly, put the phone within  $1$ – $5^\circ$  of the truth. So the phone compares the magnetometer's heading with its own and takes the difference, less the drift already measured, as the correction to where the heading started — but only from a field that has shown it is Earth's. Earth's field is fixed in the world; a field from the car, or from something carried with the phone, turns with them. Over each last minute in which the heading swept at least  $30^\circ$ , the phone expresses the field in Core Motion's world frame; if it stayed within  $8\ \mu\text{T}$  of its mean, at a strength of 25 to  $70\ \mu\text{T}$ , that minute was Earth's field and its average difference is a sample of the correction. Until the first such minute, a field between 25 and  $65\ \mu\text{T}$  whose strength and dip have been steady for a minute is used instead. Beside a MagSafe charger the phone read up to  $2,600\ \mu\text{T}$  and nothing is taken from it.

**The angle of the phone, from turns.** The heading says where the phone points, not where the vehicle goes; the two differ by an angle  $\beta$  that depends on how the phone sits. In a turn the vehicle is pushed toward the inside of the curve, with a push equal to speed times turn rate, so comparing the sideways push the phone feels with the turning the gyroscope measures shows which way is forward. Written as complex numbers,

$$\hat{\beta} = \arg \left( \frac{\sum_R z_t}{\sum_R |v_t \omega_t|} + \frac{\sum_L z_t}{\sum_L |v_t \omega_t|} \right), \quad z_t = \frac{a_t}{|a_t|} i v_t \omega_t, \quad (2)$$

where  $a_t$  is the horizontal acceleration relative to the heading, written as its sideways and forward parts,  $v_t$  the speed,  $\omega_t$  the turn rate, and  $R$  and  $L$  the seconds turning right and left. Braking into a corner and pulling away out of it lean the answer one way in right turns and the other way in left ones, so the two kinds are averaged apart. Only seconds above 14 km/h count, and only when the turn explains at least a tenth of the push; otherwise the push is braking or throttle, and a reversing car's push points backwards. The first estimate comes after 30 seconds of turning, and whenever the estimate improves the part of the ride already drawn is redrawn with it.

**Walking.** Over four seconds the back-and-forth of the steps lines up with the direction of walking, and each push-off is sharper than each landing, which tells which end of that line is forward. Steps are counted as peaks of the vertical acceleration and turned into distance with the stride Apple's pedometer has learned; the walk takes the larger of that and the pedometer's own total. In some pockets each footfall rings twice, with an echo about a quarter of a second after it, so the rhythm of the last four seconds is found from the signal's autocorrelation and, while it is clear, a peak sooner than 0.65 of its step period is refused as an echo.

### 3.3 The route

After the first point, each second's step is laid from the previous point along the current direction, for the distance the speed covers in that second. No later GPS position, course or speed enters the route, the heading or the decisions between walking and riding; the magnetic declination is taken once, at the first point. Appendix B lists every setting (Table 11).

### 3.4 Recordings

All recordings were made by the author with one iPhone, mostly in one city, with the phone's raw sensor readings saved — acceleration, rotation, orientation and, in the most recent ones, the magnetic field, 50 times a second — and GPS recorded alongside. Table 1 summarises them. A motorcycle and a car were told apart from the sensors themselves: in a turn a motorcycle leans and the phone leans with it, while a car stays level and the phone is pushed sideways; this agreed with every journey whose vehicle was known. Walking is the time the app was counting steps. One car journey was recorded with the app measuring by GPS, so the app made no walking decisions in it; it was replayed as a car and, from the first step after the car stopped, as walking, as described when recorded, with GPS from every fix the phone logged.

One flight is too few to learn the air speed from, so the two flight engines were trained on NASA's public DASHlink archive of flight recorder data [9]: the recordings of one regional jet in airline service, of which 302 flights include a takeoff. Each flight was reduced to one value a second of the recorded ground speed and of the aircraft's pitch and roll, from which its tilt from level follows as  $\arccos(\cos \theta \cos \phi)$ . The flights spent a median 48 minutes in the air (26 to 98 for the middle 80%), at a median ground speed of 656 km/h. Only the ground speed is an answer; the inputs are the four the phone forms itself, timed from the start of the takeoff roll, taken as the last moment under 30 knots before liftoff.

**Table 1.** The recordings. GPS km is the distance GPS could check; half-minutes are the stretches used for speed; direction is graded in recordings with at least 50 graded seconds.

|            | Record-ings | Hours | GPS km | Half-minutes | Direction rec. (s) | Routes | Phone                 |
|------------|-------------|-------|--------|--------------|--------------------|--------|-----------------------|
| Motorcycle | 51          | 14.2  | 223    | 1044         | 52 (16,973)        | 53     | trouser pocket        |
| Car        | 52          | 15.8  | 326    | 1326         | 43 (20,612)        | 43     | pocket, flat or mount |
| Walking    | 31          | —     | 4.2    | 103          | 7 (721)            | —      | pocket                |
| Plane      | 1           | 0.24  | 74     | —            | 1 (876)            | —      | cabin                 |

### 3.5 Measuring the newest version

So that every result describes the newest version, each recording was replayed through the current method from its raw sensor data:

- **Speed:** the same fingerprint, the same 12-neighbour average and correction, refusals answered by a starting network trained without that recording, and the same rules for stops and a phone in the hand. The store holds 4,000 examples drawn at random from every *other* journey, so no journey is measured with its own GPS in the store. Each example keeps the same chance of being drawn whatever recordings are added, so a new recording changes the other journeys' stores only by its own examples. On 12 recent journeys the replay differed from what the app itself had drawn by a median 7%.
- **Direction:** the heading reference, drift correction, magnetometer correction (where the magnetic field was recorded), the angle learned from turns and the redrawing, with the replayed speed.
- **Starting network:** each recording measured by a network trained without it (the recordings split into eight groups), and the store tested holding only 60 to 4,000 examples.
- **Walking:** the step detector replayed on the recorded acceleration with the stride the phone had learned.
- **The flight:** the app's own code for the takeoff integration, the tilt and both flight engines, compiled and run on the flight's raw samples; its direction is replayed with that speed in the air, as the app has it.
- **Flight engines on NASA flights:** each flight measured by engines trained without it (the flights split into five groups), from the start of the roll to landing, against holding the speed two minutes into the roll.

Three inputs are taken from the recordings as they were: whether each second was walking or riding, Apple's motion classifier, and the car-park ramp flag. Cabin pressure was not recorded, so on the flight the moment the app would call it airborne was taken from the same integration's climb rate.

In a pocket GPS is itself accurate only to 10–40 m, so it is used over stretches long enough for that not to matter.

- **Speed and distance** are compared over half-minute stretches between GPS fixes better than 40 m: the straight distance GPS moved against the distance the system added. For walking, only straight stretches between fixes better than 25 m are used.
- **Direction** is compared every second: the bearing from the GPS position 15 s earlier to the one 15 s later, against the system's heading over the same half-minute (10 s for the plane).

- **Whole routes** are compared by finding the rotation and the size change that best lay the drawn route over the GPS track, pinned at the shared start, for routes of at least 1 km. The rotation is how far the route is turned; the size change how much too big or small it is; the drift is how far the drawn end lies from GPS's, as a share of the distance travelled.

10 car journeys went into, started in or ended in underground car parks. There GPS kept claiming 10 m accuracy for positions it did not have — frozen while parked, wandering on the ramps — so those stretches, found from the barometer, are left out of every comparison. So are half-minutes that mix walking and riding, and those in which the app counted steps while GPS showed riding speed, and the 47 seconds of one journey in which iOS had ended the app.

## 4 Results by kind of travel

Each kind of travel is shown on its own and measured the same way (Section 3.5): speed over half-minutes, distance over journeys, direction every second, and whole routes. Section 6 then sets them side by side.

### 4.1 Motorcycle

Figure 2 and Table 2 show the motorcycle, with the phone in a trouser pocket.

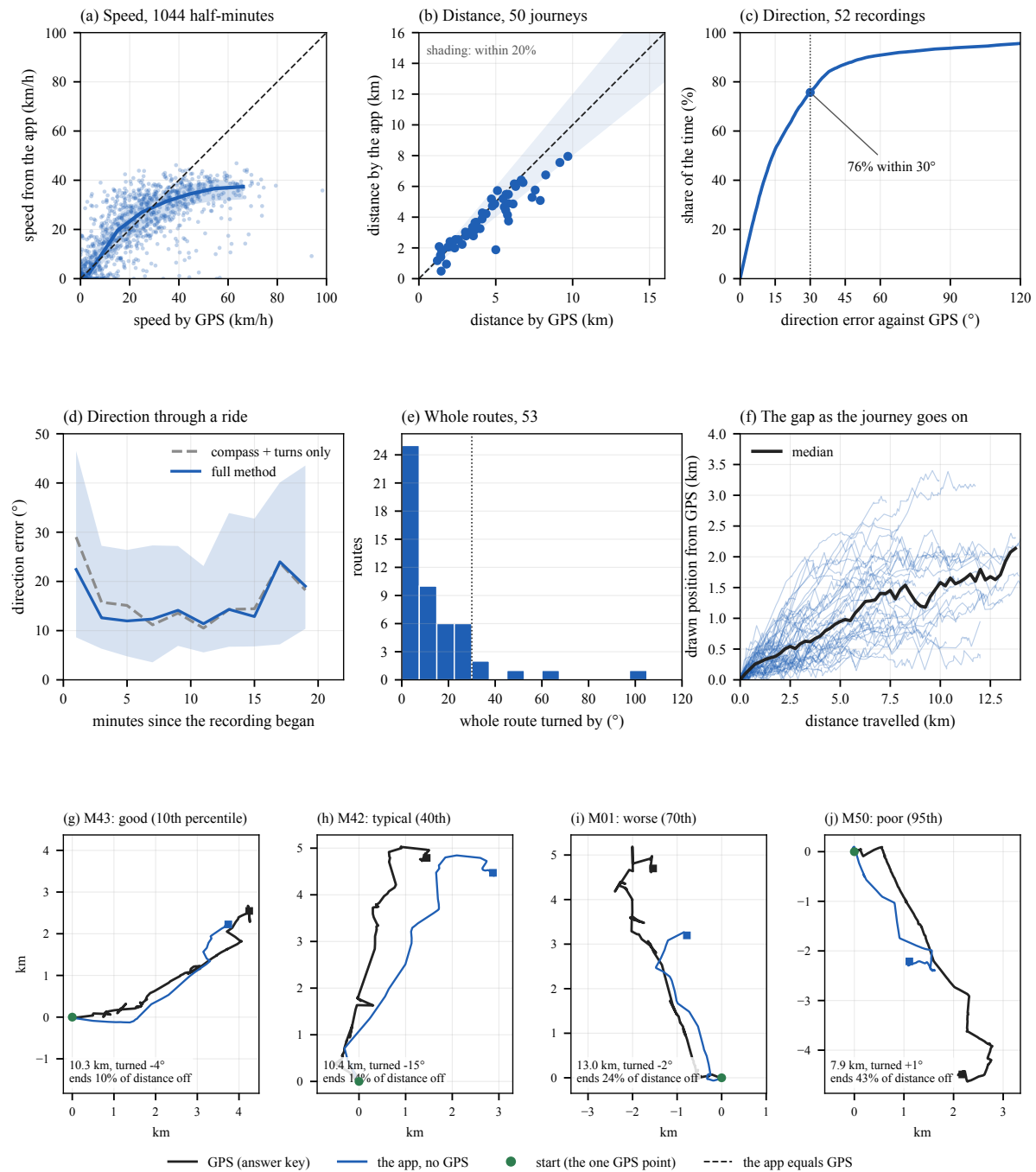
**Table 2.** Motorcycle at a glance (left), and speed by GPS speed band (right): the number of half-minutes, the middle GPS and app speeds, and the average error.

| <i>Recordings</i>                  |                        | GPS band |          | Middle (km/h) |      | Error  |
|------------------------------------|------------------------|----------|----------|---------------|------|--------|
|                                    |                        | (km/h)   | <i>n</i> | GPS           | App  | (km/h) |
| Journeys, hours                    | 51, 14.2               | 0–10     | 320      | 2.5           | 2.1  | 4.5    |
| Checked by GPS                     | 223 km                 | 10–20    | 204      | 15.2          | 19.6 | 8.1    |
| Phone                              | trouser pocket         | 20–30    | 176      | 25.0          | 26.9 | 7.4    |
| <i>Speed and distance</i>          |                        | 30–40    | 138      | 35.0          | 31.4 | 7.2    |
| Half-minutes graded                | 1,044                  | 40–50    | 107      | 44.0          | 34.4 | 12.3   |
| Average speed error                | 9.0 km/h               | 50–60    | 54       | 54.0          | 36.6 | 20.6   |
| Distance, all journeys             | –12.4% (195 of 223 km) | 60–80    | 42       | 65.3          | 37.4 | 32.0   |
| Median journey                     | –5.2%                  | 80+      | 3        | 98.4          | 35.8 | 74.3   |
| Journeys within 10%, 20%           | 50%, 72%               | All      | 1044     | 20.0          | 21.5 | 9.0    |
| Range, app/GPS                     | 0.34–1.58              |          |          |               |      |        |
| <i>Direction</i>                   |                        |          |          |               |      |        |
| Graded                             | 52 rec., 16,973 s      |          |          |               |      |        |
| Median error                       | 14°                    |          |          |               |      |        |
| 75th, 90th percentile              | 29°, 54°               |          |          |               |      |        |
| Within 15°, 30°, 45°               | 53%, 76%, 87%          |          |          |               |      |        |
| <i>Whole routes, at least 1 km</i> |                        |          |          |               |      |        |
| Routes                             | 53, 433 km             |          |          |               |      |        |
| Turned, median                     | 9°                     |          |          |               |      |        |
| Within 15°, 30°                    | 66%, 89%               |          |          |               |      |        |
| Turned more than 45°               | 4                      |          |          |               |      |        |
| Size, median                       | 0.80 (47% within 20%)  |          |          |               |      |        |
| End drift, median                  | 17% of distance        |          |          |               |      |        |

**Speed.** The app squeezes the range: 10–20 km/h reads about 20, 20–30 km/h about 27 where GPS measured 25, and above that it reads ever lower: about 31 at 30–40 km/h and about 37 where GPS measured 65 at 60–80 km/h. The squeeze comes from the averaging: when a fingerprint could belong to several speeds, the average lands in the middle. The average error is 9.0 km/h over 1,044 half-minutes.

**Distance.** The app added 195 km where GPS measured 223 km, 12% short: the distance lost at high speed outweighs what is gained at low speed. Journey by journey the median was 5.2% short, and 72% of journeys came within 20% of GPS.

**Direction.** Over 52 recordings the median error is 14° and 76% of seconds are within 30°. The tail is long: one second in ten is off by more than 54°, because a phone in a trouser pocket moves as the rider shifts. In the first minutes the angle of the phone is still being learned and the error is highest; it settles within about five minutes. After that the full method and the compass heading with the angle from turns alone are close, since a motorcycle's stops rarely hold the phone still enough to measure drift.



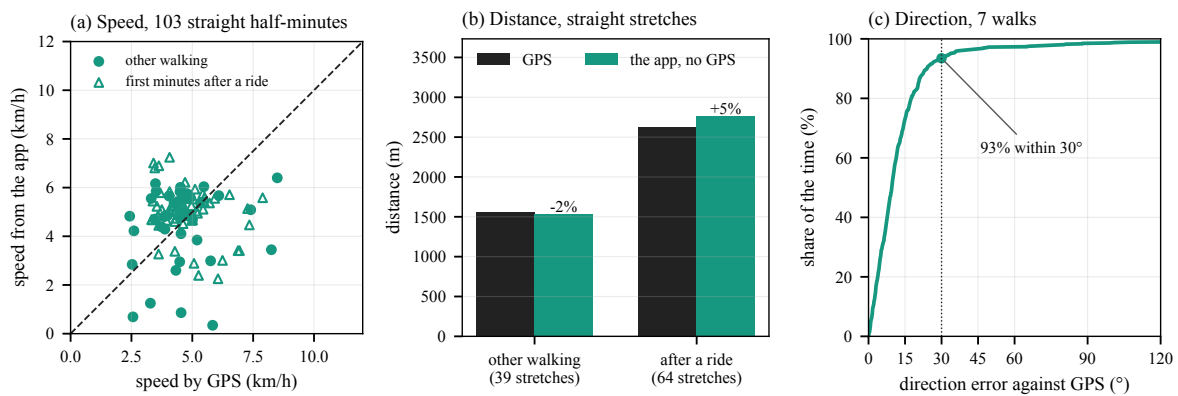
**Figure 2.** Motorcyclist, with the phone in a trouser pocket. (a) Speed per half-minute against GPS; the line is the middle of each 10 km/h band, the shading the middle half, and on the dashed line the two agree. (b) Distance per journey of at least 1 km; the shading is within 20%. (c) How often the direction is within a given angle of GPS. (d) Direction error through a ride, in two-minute bins with at least 300 graded seconds: the full method against the compass heading with the angle from turns alone. (e) How far each whole route is turned. (f) How far the drawn position is from GPS as the journey goes on: one thin line per route, and the median. (g–j) Four routes at the 10th, 40th, 70th and 95th percentile of end drift, labelled as in Appendix A; each is turned together with its GPS track by an arbitrary angle so as not to show where it was recorded.

**Whole routes.** The median route is turned  $9^\circ$  and 89% are within  $30^\circ$ . Routes come out small — a median 0.80 of the real size — partly because distance is short and partly because small direction errors fold the path, and the drawn end lies a median 17% of the distance travelled from GPS's. Routes turned more than  $45^\circ$ : 4, of them 3 under 3 km, with too few turns to learn the angle of the phone well.

## 4.2 Walking

Walking is counted by steps, with the stride the phone has learned (Figure 3, Table 3). Over 103 straight half-minutes the app read about 5.1 km/h where GPS measured 4.6, and counted 3% more distance than GPS: 5% more in the first minutes after a ride and 2% less at other times. Direction comes from the back-and-forth of the steps: over 7 longer walks the median error is 10° and 93% of seconds are within 30°.

On 33 of the 103 stretches the steps counted ran more than 15% faster than the walking rhythm itself; on the walk after one motorcycle ride the count was 2.5 to 2.75 steps a second against a rhythm of 1.85, and those stretches came out close to twice what GPS saw. Refusing the echo removes most of the extra steps. What that does to distance cannot be re-run: the app takes the larger of its own count and Apple’s pedometer, which was not recorded until the newest version, so the re-run models the pedometer from what each version logged, and its walking figures above do not move. Scaled by the change in the count alone, the walking total would go from +2.6% to about −4.5% and the stretches within 20% from 43% to 55%; 44 stretches come closer to GPS and 10 move away, all but one of them already short, and that one within 1% of GPS. That is an estimate until walks recorded with the pedometer arrive. The first three had no echo: the app’s count matched the pedometer’s and the walking rhythm, and the check refused almost nothing. On the one with usable GPS the distance still came out long, by the stride rather than the count: the stride the pedometer had learned was longer than the steps of that slow walk uphill.



**Figure 3.** Walking, with the phone in a pocket. (a) Speed over straight half-minutes against GPS; triangles are the first minutes after a ride. (b) Distance over the straight stretches, added up. (c) How often the direction is within a given angle of GPS, over 7 longer walks.

**Table 3.** Walking: straight half-minute stretches between GPS fixes better than 25 m.

| Straight stretches         | <i>n</i> | Distance (m) |      |            | Speed (km/h) |     |
|----------------------------|----------|--------------|------|------------|--------------|-----|
|                            |          | GPS          | App  | Difference | GPS          | App |
| Other walking              | 39       | 1555         | 1530 | −1.6%      | 4.5          | 4.8 |
| First minutes after a ride | 64       | 2630         | 2765 | +5.1%      | 4.7          | 5.1 |
| All                        | 103      | 4185         | 4296 | +2.6%      | 4.6          | 5.1 |

### 4.3 Car

Figure 4 and Table 4 show the car, with the phone in a pocket, lying flat or in a mount.

**Table 4.** Car at a glance (left), and speed by GPS speed band (right): the number of half-minutes, the middle GPS and app speeds, and the average error.

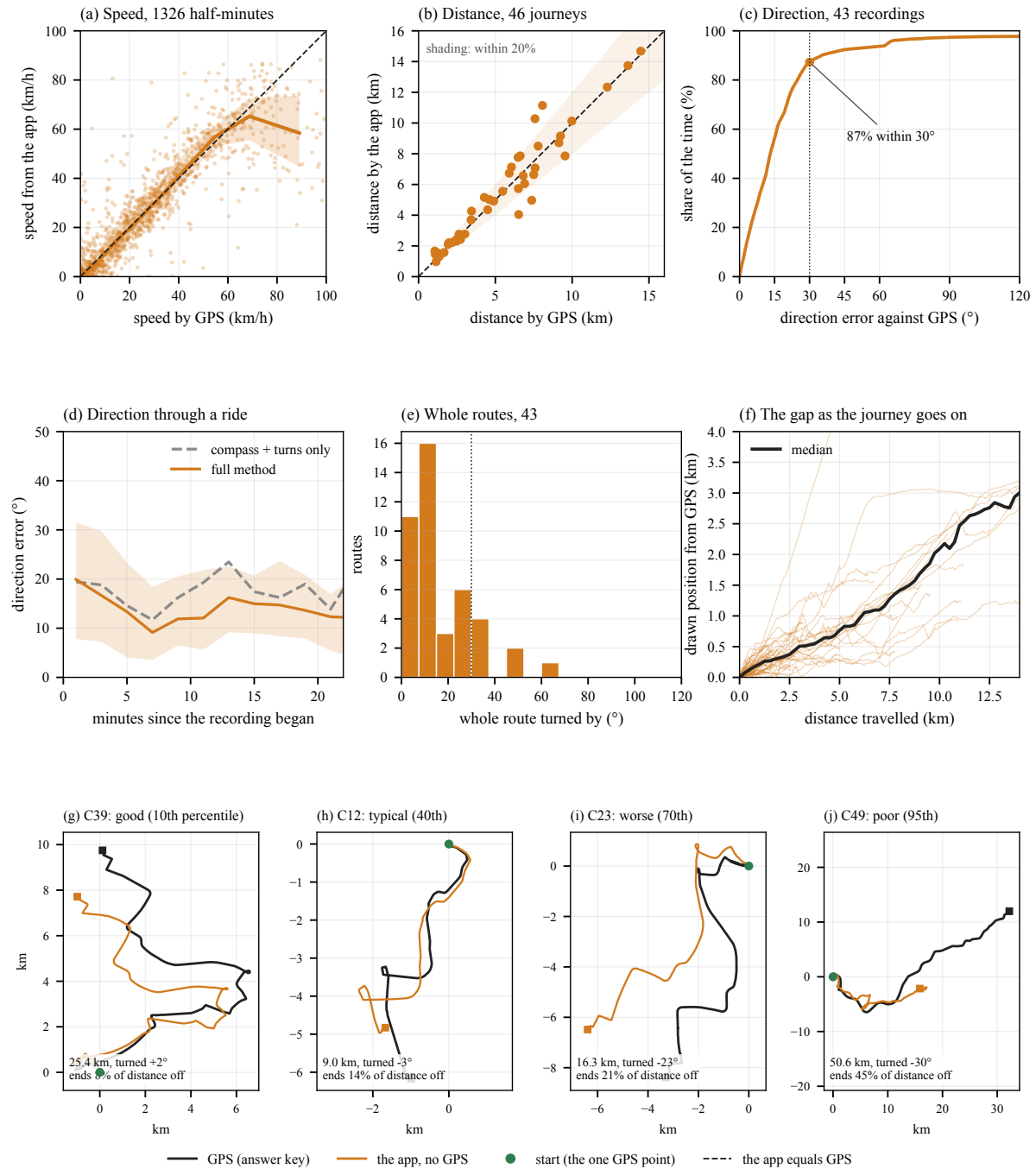
| <i>Recordings</i>                  |                       | GPS band |          | Middle (km/h) |      | Error  |
|------------------------------------|-----------------------|----------|----------|---------------|------|--------|
|                                    |                       | (km/h)   | <i>n</i> | GPS           | App  | (km/h) |
| Journeys, hours                    | 52, 15.8              | 0–10     | 369      | 2.7           | 2.5  | 3.4    |
| Checked by GPS                     | 326 km                | 10–20    | 227      | 15.0          | 14.9 | 6.1    |
| Phone                              | pocket, flat or mount | 20–30    | 201      | 25.2          | 24.7 | 5.5    |
| <i>Speed and distance</i>          |                       | 30–40    | 168      | 35.0          | 35.1 | 5.9    |
| Half-minutes graded                | 1,326                 | 40–50    | 108      | 44.4          | 45.7 | 7.0    |
| Average speed error                | 6.9 km/h              | 50–60    | 89       | 54.3          | 56.8 | 7.0    |
| Distance, all journeys             | −2.6% (317 of 326 km) | 60–80    | 116      | 68.5          | 65.2 | 10.4   |
| Median journey                     | +1.0%                 | 80+      | 48       | 89.0          | 57.0 | 38.7   |
| Journeys within 10%, 20%           | 54%, 78%              | All      | 1326     | 23.5          | 24.3 | 6.9    |
| Range, app/GPS                     | 0.62–1.57             |          |          |               |      |        |
| <i>Direction</i>                   |                       |          |          |               |      |        |
| Graded                             | 43 rec., 20,612 s     |          |          |               |      |        |
| Median error                       | 13°                   |          |          |               |      |        |
| 75th, 90th percentile              | 22°, 35°              |          |          |               |      |        |
| Within 15°, 30°, 45°               | 56%, 87%, 92%         |          |          |               |      |        |
| <i>Whole routes, at least 1 km</i> |                       |          |          |               |      |        |
| Routes                             | 43, 334 km            |          |          |               |      |        |
| Turned, median                     | 13°                   |          |          |               |      |        |
| Within 15°, 30°                    | 63%, 84%              |          |          |               |      |        |
| Turned more than 45°               | 3                     |          |          |               |      |        |
| Size, median                       | 0.97 (84% within 20%) |          |          |               |      |        |
| End drift, median                  | 17% of distance       |          |          |               |      |        |

**Speed.** Much closer than on the motorcycle: from 10 to 60 km/h the middle reading is within about 3 km/h of GPS. Above that it reads low: 65 where GPS measured 69, and 57 where it measured 89. The average error is 6.9 km/h over 1,326 half-minutes.

**Distance.** The app added 317 km where GPS measured 326 km, 2.6% short. Journey by journey the median was 1.0% long, and 78% of journeys came within 20% of GPS.

**Direction.** Over 43 recordings the median error is 13° and 87% of seconds are within 30°; one second in ten is off by more than 35°. The full method stays below the compass heading with the angle from turns alone for most of a drive, as drift measured at stops and the magnetometer keep the heading reference in place.

**Whole routes.** The median route is turned 13° and 84% are within 30°. Car routes have the right size (median 0.97), and the drawn end lies a median 17% of the distance travelled from GPS's. Routes turned more than 45°: 3, of them 2 under 3 km.



**Figure 4.** Car, with the phone in a pocket, lying flat or in a mount. (a) Speed per half-minute against GPS; the line is the middle of each 10 km/h band, the shading the middle half, and on the dashed line the two agree. (b) Distance per journey of at least 1 km; the shading is within 20%. (c) How often the direction is within a given angle of GPS. (d) Direction error through a ride, in two-minute bins with at least 300 graded seconds: the full method against the compass heading with the angle from turns alone. (e) How far each whole route is turned. (f) How far the drawn position is from GPS as the journey goes on: one thin line per route, and the median. (g–j) Four routes at the 10th, 40th, 70th and 95th percentile of end drift, labelled as in Appendix A; each is turned together with its GPS track by an arbitrary angle so as not to show where it was recorded.

## 4.4 Plane

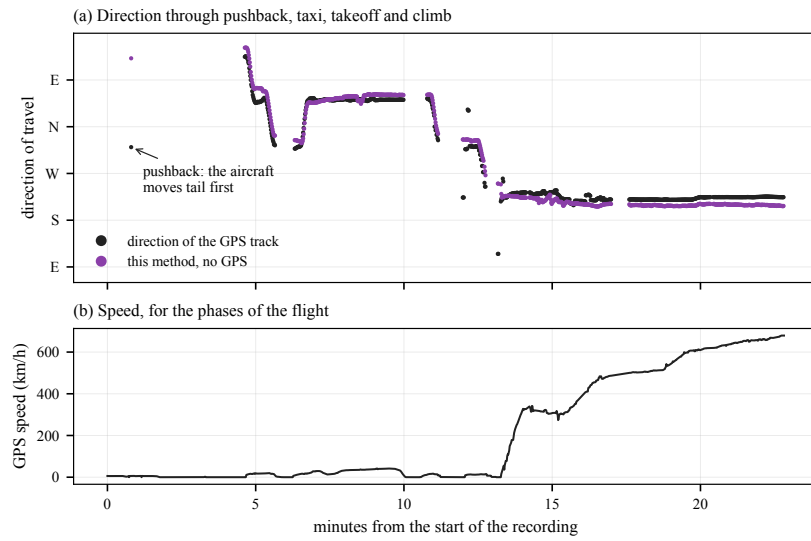
On the ground the plane is measured like a car: while taxiing the app reads about 12 km/h where GPS measured 18. In the air a smooth cabin reads as standing still, so the phone measures the takeoff — 265 km/h forty seconds into the roll, where GPS measured 266 — and after two minutes the flight network takes over, reading about 570 km/h where GPS measured 510 three minutes later and 675 where it measured 670 near the end.

Speed in the air is where the flight engines come in (Figure 6, Table 5). On the recorded flight the takeoff measurement reached 317 km/h and, held, counted 45.2 of the 74.1 km: the aircraft kept speeding up to 680 km/h. From two minutes into the roll the flight network read the climb from the phone's tilt and the time since the roll, and the flight counted 76.3 km (103%); with the flight store, 73.7 km (99%). The average speed error in the air fell from 196 km/h to 38 and 37. The network reads high by about 50 km/h early in the climb, where this aircraft gained speed more slowly than the NASA jet typically did, and matches GPS near the end.

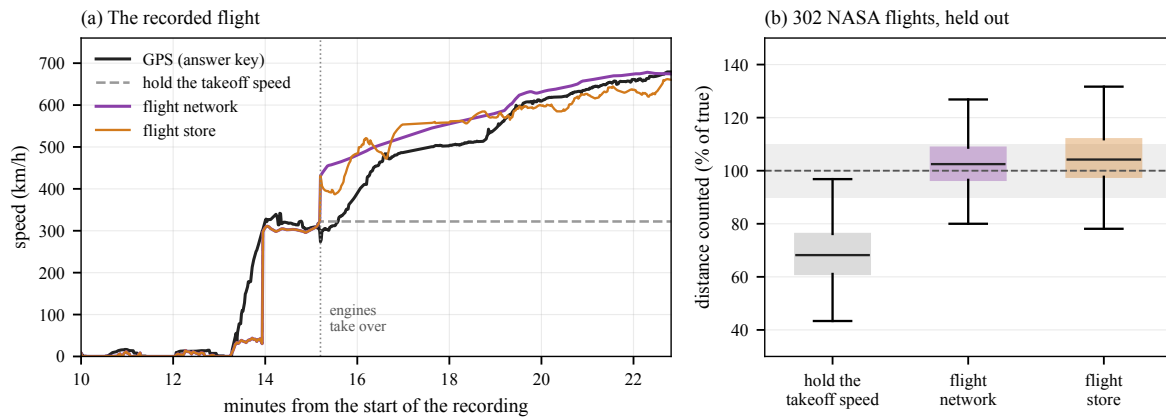
On NASA's flights, each measured without its own data, holding the takeoff speed counted a median 68% of the distance and came within 10% on 7% of flights. The network counted a median 102% and came within 10% on 68% of flights; the store 104% and 65%. The two engines agree closely; the network, being smaller to ship and a little closer, is the one the app uses.

Figure 5 follows the flight with the same direction method as on the road, fed the speed the app itself has: the speed model on the ground, then the measured takeoff and the flight network. While taxiing the median error was 9° and in the air 12°. In all, 12 seconds were more than 30° off: 1 in the pushback, when the aircraft moves tail first while the method gives the way the nose points, and 11 around the runway, stopped or just starting the roll, where the GPS track has no clear direction. In the climb the error grows slowly, to about 15° by the end of the recording.

Figure 7 puts speed and direction together: the whole flight drawn from the first GPS point with nothing but what the phone senses. Both versions stay within about a kilometre of GPS through the taxi and the takeoff, a median 1.0 km over the whole recording. In the air the direction settles about 12° to one side of the track, and that, more than the speed, carries the drawn ends 16.9 km (network) and 16.3 km (store) from where GPS ended, 23% of the distance flown. The two versions draw nearly the same route: they share the direction and differ in speed by 3.5%.



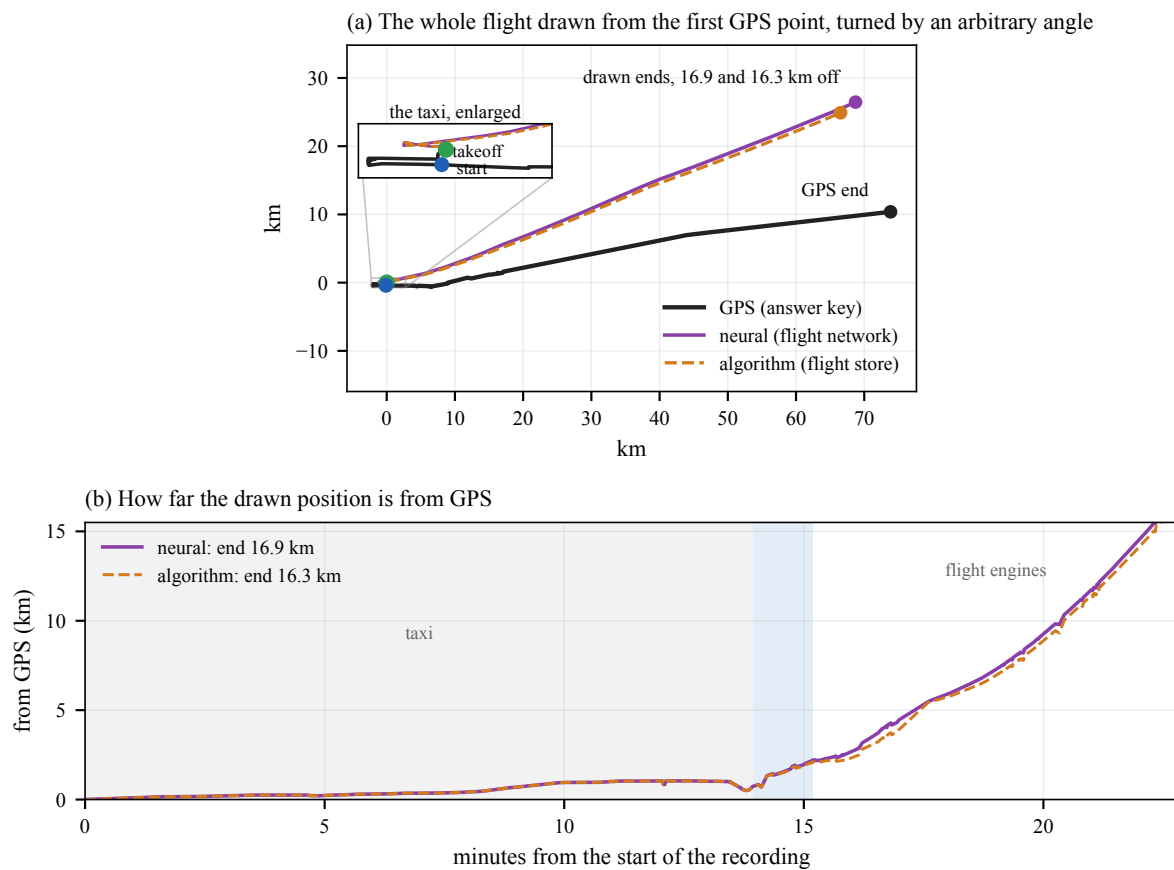
**Figure 5.** The flight, from the gate to the climb. (a) Direction of travel from the GPS track (black) and from this method, which uses no GPS (purple). (b) The aircraft's speed from GPS.



**Figure 6.** The flight engines. (a) The recorded flight: the takeoff is measured for two minutes, after which holding its speed (dashed) misses the climb while both engines follow it. (b) NASA flights, each measured by engines trained without it: the share of the distance from the start of the roll to landing that each counted; the box is the middle half, the whiskers the rest without outliers, the band within 10%.

**Table 5.** Speed in the air. Left: NASA flights, each measured by engines trained without it, from the start of the roll to landing. Right: the phone's flight through the app's own code. Speed errors are average km/h in the air (per flight, the median over flights).

| In the air             | 302 NASA flights, held out |                |            |             | The phone's flight |                  |             |
|------------------------|----------------------------|----------------|------------|-------------|--------------------|------------------|-------------|
|                        | Median distance            | 10th–90th pct. | Within 10% | Speed error | Distance km        | Share of 74.1 km | Speed error |
| Hold the takeoff speed | 68%                        | 55–88%         | 7%         | 225         | 45.2               | 61%              | 196         |
| Flight network         | 102%                       | 91–118%        | 68%        | 82          | 76.3               | 103%             | 38          |
| Flight store           | 104%                       | 92–122%        | 65%        | 88          | 73.7               | 99%              | 37          |



**Figure 7.** The whole flight drawn with no GPS after the first point. (a) GPS (black), the flight network (purple) and the flight store (orange, dashed). Like the example routes in Figures 2 and 4 it is turned by an arbitrary angle so as not to show where it was recorded; the inset enlarges the taxi. (b) The distance between each drawn position and GPS's; grey is the taxi, blue the measured takeoff.

## 5 A new phone: the built-in networks

A phone that has learned nothing yet still has to answer. Two small networks ship with the app for that: one reads speed from the vibration fingerprint on the ground, the other gives the speed in the air after the measured takeoff. Figure 8 draws both.

(a) Speed network, for a phone with fewer than 3,000 examples



(b) Flight network, in the air after the measured takeoff

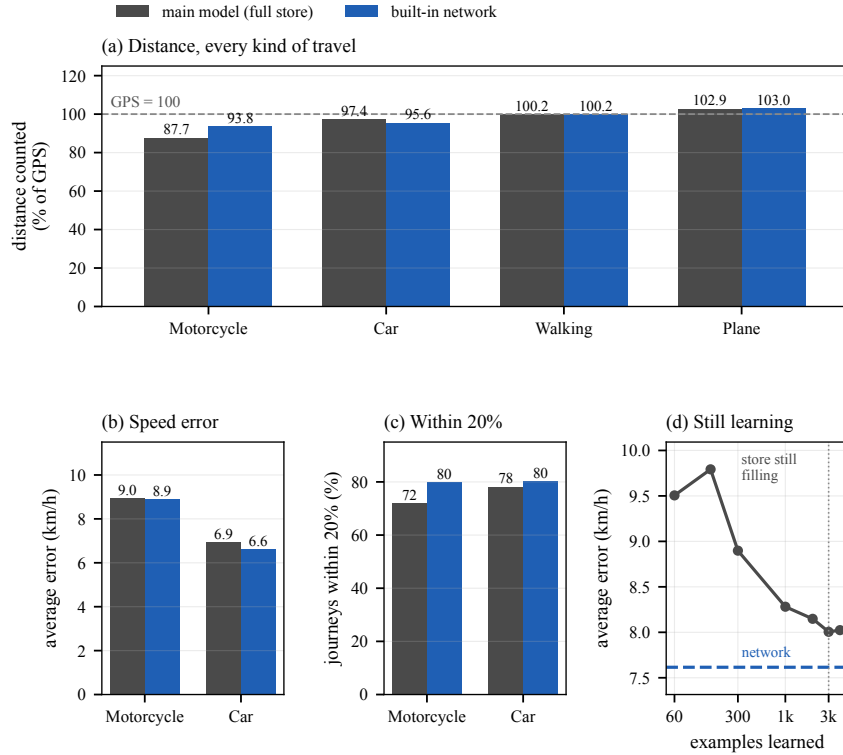


**Figure 8.** The two built-in networks, drawn layer by layer; each hidden layer has 32 units, of which only some are drawn. (a) The speed network: the fingerprint’s eleven numbers go through two layers of 32 and one output, which a calibration then stretches; a fingerprint unlike any it was trained on gets no answer. (b) The flight network: four numbers from the phone’s tilt and the time since the takeoff roll go through the same two layers, and the output is a ground speed in hundreds of km/h.

### 5.1 How the speed network works

Its inputs are the fingerprint of Section 3.1: the energy of the up-and-down shaking in nine frequency bands from 0.4 to 24 Hz, and two measures of its overall strength — its spread, and how sharply it changes from one sample to the next. Each is standardised with a mean and scale fixed from the training data, never from the phone’s own store, so clearing the store cannot change the network. Two layers of 32 units with ReLU activations follow, then one output unit with a softplus, which keeps the answer above zero: 1,473 numbers in all, evaluated once a second.

An averaging model pulls its answers toward the middle, so the output goes through a calibration, as the store’s does: the network’s answers on its training data at 21 quantiles are matched to the true speeds at the same quantiles, and each new answer is mapped along that line. And like the store it may refuse: the training fingerprints were grouped into 32 clusters, and a fingerprint farther from every cluster centre than 99.5% of the training fingerprints gets no answer rather than a guess.



**Figure 9.** The built-in network (blue) next to the main model with a full store (grey), on the same journeys, the network only on recordings it had never seen. (a) Share of the GPS distance counted. (b) Average speed error. (c) Journeys within 20%. (d) A store still filling up; the dotted line is where the app hands over.

It was trained on the GPS-labelled fingerprints of the recorded trips, with a Huber loss (which weighs large errors less than a squared loss does), the Adam optimiser and 40 passes over the data. The app uses it until the phone’s own store holds 3,000 examples. For the results below each recording is measured by a network trained without it, the recordings split into eight groups.

## 5.2 How the flight network works

In the air the fingerprint says nothing, since a smooth cabin reads as standing still, so the flight network’s four inputs come from the phone’s tilt instead (Section 3.1): minutes since the takeoff roll began, the average tilt over the last minute and the last five minutes, and the tilt added up since the roll. Inputs beyond the range of the training flights are held at its edge, then standardised; two layers of 32 ReLU units follow, and the single output is the ground speed in hundreds of km/h: 1,249 numbers. It was trained on every third second of the 302 NASA flights (Section 3.4) with the Adam optimiser, stopping when a held-back tenth of the data stopped improving. The flight store beside it holds 4,000 of the same examples and answers with the 12 nearest. How both do on NASA’s flights and on the recorded one is in Section 4.4.

## 5.3 How well the speed network works

A new phone starts with the built-in network. On recordings it had never seen it missed by 8.9 km/h on a motorcycle and 6.6 km/h in a car, against 9.0 and 6.9 for the main model with a full store, and counted 93.8% and 95.6% of the distance against 87.7% and 97.4% (Figure 9).

**Table 6.** A store still filling up against the built-in network, over motorcycle and car journeys together, with each journey’s own examples left out.

| Speed model           | Average speed error (km/h) |            |     | Journeys |            |
|-----------------------|----------------------------|------------|-----|----------|------------|
|                       | All                        | Motorcycle | Car | Distance | within 20% |
| Store, 60 examples    | 9.5                        | 9.7        | 9.4 | −3.2%    | 61%        |
| Store, 150 examples   | 9.8                        | 10.3       | 9.4 | −9.0%    | 54%        |
| Store, 300 examples   | 8.9                        | 9.6        | 8.3 | −6.4%    | 71%        |
| Store, 1,000 examples | 8.3                        | 9.1        | 7.6 | −7.7%    | 71%        |
| Store, 2,000 examples | 8.1                        | 9.0        | 7.4 | −7.5%    | 75%        |
| Store, 3,000 examples | 8.0                        | 9.0        | 7.2 | −7.6%    | 75%        |
| Store, 4,000 examples | 8.0                        | 9.2        | 7.1 | −8.1%    | 68%        |
| Built-in network      | 7.6                        | 8.9        | 6.6 | −5.1%    | 80%        |

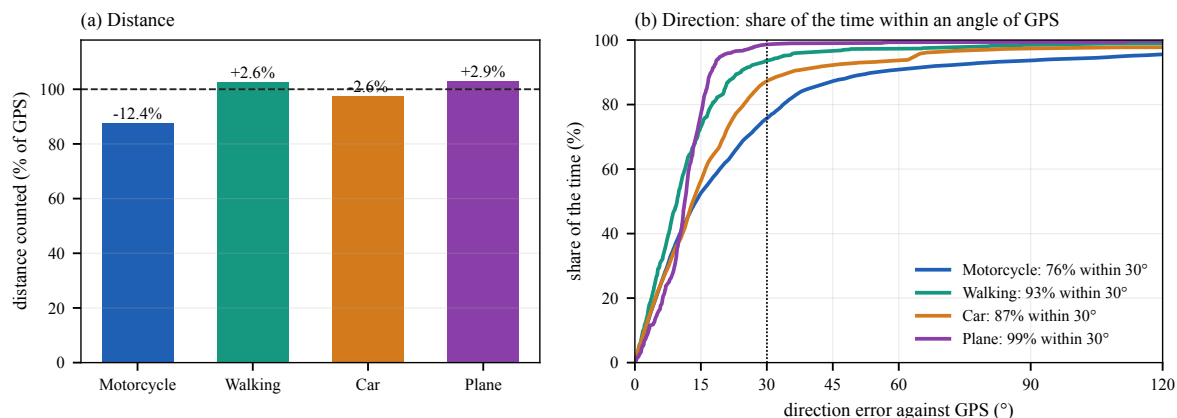
A store still filling up does worse (Table 6): 9.5 km/h with 60 examples, 8.3 with 1,000 and 8.0 with 4,000, still a little behind the network’s 7.6. The app hands over to the store at 3,000 examples all the same, because the store keeps adapting to the phone’s own vehicle and pocket, an advantage these recordings, made with the same few vehicles the network was trained on, cannot show. Walking (100.2%) and the flight (102.9% and 103.0%) are nearly the same with either, since they are counted by steps and, in the air, by the flight engines.

## 6 Results across every kind of travel

Table 7 and Figure 10 set the four kinds of travel side by side. Direction is the stronger half everywhere: within  $30^\circ$  of GPS 76% of the time on a motorcycle, 87% in a car, 93% on foot and 99% on the plane. Distance is close in a car ( $-2.6\%$ ), on foot ( $+2.6\%$ ) and on the plane ( $+2.9\%$ ), and short on a motorcycle ( $-12.4\%$ ), where the speed model squeezes the range. Motorcycle and car together, the app added 512 km where GPS measured 548 km, 6.5% short.

**Table 7.** Every kind of travel side by side.

|                             | Motorcycle       | Walking                | Car               | Plane                    |
|-----------------------------|------------------|------------------------|-------------------|--------------------------|
| Recorded                    | 51 journeys      | 103 stretches          | 52 journeys       | 1 flight                 |
| GPS km checked              | 223              | 4.2                    | 326               | 74                       |
| Distance against GPS        | $-12.4\%$        | $+2.6\%$               | $-2.6\%$          | $+2.9\%$                 |
| Speed                       | error 9.0 km/h   | 5.1 for GPS's 4.6 km/h | error 6.9 km/h    | error 38 km/h in the air |
| Direction, median error     | $14^\circ$       | $10^\circ$             | $13^\circ$        | $11^\circ$               |
| Direction within $30^\circ$ | 76%              | 93%                    | 87%               | 99%                      |
| Whole route, median         | turned $9^\circ$ | —                      | turned $13^\circ$ | ends 16.9 km off         |



**Figure 10.** Every kind of travel side by side. (a) Distance counted, as a share of GPS's. (b) How often the direction is within a given angle of GPS.

### 6.1 What each part of the direction method adds

Table 8 replays every recording with the direction method built up one part at a time, graded exactly as above. For cars the share of seconds within  $30^\circ$  goes from 67% with the heading and the angle from turns alone to 70% with drift measured at stops, 73% with right and left turns averaged apart, 73% with the push check, 76% with the clean magnetometer, 80% when drift is measured only in Earth's field and 87% when north is taken only from a field checked by turning; routes within  $30^\circ$  go from 67% to 84%, and routes of either vehicle turned more than  $45^\circ$  from 9 to 7. Drift measured at stops helps a drive where the phone sat in an Earth-like field and harms one where the field was disturbed (Section 7); counting only stops in Earth's field keeps the first and removes the second. For motorcycles every version is within 1 points of the others. The flight, replayed with the speed the app has in the air, is within  $30^\circ$  83% of the time with the first version and 99% once drift is measured at stops.

**Table 8.** What each part of the direction method adds: the share of graded seconds and of whole routes within  $30^\circ$ , routes turned more than  $45^\circ$ , and the flight's seconds within  $30^\circ$ . Each row adds one part to the row above. The magnetometer applies only where the magnetic field was recorded (19 car drives and 12 motorcycle rides so far).

| Direction method              | Motorcycle |        | Car     |        | Routes          | Flight  |
|-------------------------------|------------|--------|---------|--------|-----------------|---------|
|                               | Seconds    | Routes | Seconds | Routes | over $45^\circ$ | seconds |
| Heading + angle from turns    | 76%        | 87%    | 67%     | 67%    | 9               | 83%     |
| + drift at stops              | 75%        | 87%    | 70%     | 72%    | 10              | 99%     |
| + right, left apart           | 76%        | 89%    | 73%     | 74%    | 10              | 99%     |
| + turn explains push          | 76%        | 89%    | 73%     | 79%    | 10              | 99%     |
| + clean magnetometer          | 76%        | 89%    | 76%     | 79%    | 10              | 99%     |
| + drift only in Earth's field | 76%        | 89%    | 80%     | 81%    | 10              | 99%     |
| + north checked by turning    | 76%        | 89%    | 87%     | 84%    | 7               | 99%     |

## 7 Discussion

### 7.1 Where it goes wrong

- **On a motorcycle, fast riding reads slow and slow riding reads fast.** Above 60 km/h the speed is under two-thirds of the true value (Table 2). Over a journey the two partly cancel, but a journey that is mostly fast comes out short.
- **A constant turn over a whole drive.** Much of the remaining direction error is one angle that lasts the whole drive: 30 of the 95 graded motorcycle and car recordings are turned by more than  $15^\circ$  on average. Removing one such turn per drive — which needs GPS, so it only bounds what is possible — would raise the share of seconds within  $30^\circ$  from 82% to 89% and cut the median error from  $13.4^\circ$  to  $8.0^\circ$ . It comes from where the heading started and from the angle learned for the phone, and the replay cannot separate the two. On the car drive with a clean magnetic field the magnetometer cut it from  $29^\circ$  to  $18^\circ$ , and the seconds within  $30^\circ$  rose from 49% to 97%.
- **The first minute of a ride.** Until the angle of the phone is learned the direction comes from the heading alone (Figures 2d and 4d). The redrawing fixes the saved route afterwards, but a very short ride may never learn the angle well.
- **A short, slow drive.** The angle is learned only while the speed model reads more than 14 km/h, so on a short drive in traffic it rests on a few turns, and which turns depends on the speed model. On one car drive, ten minutes at a median 9 km/h, the speeds the app read on the day kept 100% of the 87 graded seconds within  $30^\circ$  of GPS. Replayed with the speed model rebuilt from the other journeys, a different handful of turns sets the angle each time that model changes: over the last 5 re-runs, which differed only in the journeys added, between 26% and 43% of those seconds were within  $30^\circ$ .
- **A phone that moves.** A hand shakes the phone and shaking reads as speed. A hand can no longer raise the speed, but a phone picked up while slowing keeps the speed it had when it was picked up. A phone that shifts in the pocket turns the drawn route by about as much as it moved, until enough new turns have been learned.
- **A store full of another vehicle.** The store keeps 4,000 examples and each new one replaces a random old one, so a run of car drives leaves it mostly car. On the next two motorcycle rides it declined 41–44% of the riding seconds, and the app, which then held its last answer, read 49% and 58% of GPS as recorded. The newest version answers those seconds with the starting network, which on the same seconds reads 89% and 88%. The

re-run draws its store from every journey, so it cannot show the forgetting itself.

- **A disturbed field at stops.** On one 49-minute drive the field read about  $80\ \mu\text{T}$  at stops and 165 at speed. The stops then measured a drift that did not happen while moving, and the route turned further off every minute: with every still second counted, 44% of seconds were within  $30^\circ$ ; with no drift at all, 85%; counting only stops in Earth's field, 94%. The bound matters: allowing up to  $65\ \mu\text{T}$  let the bad stops, at 62 to 65, back in. Only the recordings with the magnetometer logged, 32 so far, can show this at all, and the gain rests on that one drive.
- **A field that looks like Earth's but is not.** On the same drive the steady-field test found seconds of Earth-like strength, steady for a minute, and took the correction from them; GPS shows it was about  $30^\circ$  wrong, and those seconds dipped  $14\text{--}28^\circ$  where Earth's field here dips about  $35^\circ$ . A steady field proves little while a car stands or runs straight. Checked sample by sample against GPS on the 32 recordings with the magnetometer logged, the minutes that passed the turn check put north within  $15^\circ$  of the truth 83% of the time (median error  $7^\circ$ , worst  $68^\circ$ , 1078 samples); the steady-field seconds did so 65% of the time (median  $11^\circ$ , worst  $171^\circ$ , 1760 samples). Over those recordings' graded seconds the turn check moved the share within  $30^\circ$  from 80% to 89%. Matching the field's strength and dip to Earth's values instead was fragile (Table 9).
- **A magnet beside the phone.** By a car's MagSafe charger the phone read up to  $2,600\ \mu\text{T}$ , fifty times Earth's field, yet reported its compass as well calibrated; the iPhone's own Compass then drifted  $49^\circ$  with the gyroscope. The app ignores such a field. On the drive with the charger beside the phone throughout, a few turning minutes still showed Earth's field, and with them the seconds within  $30^\circ$  rose from 0% to 43%; a phone that never leaves such a field cannot be corrected at all.
- **Car-park ramps and rides never recognised.** A car crawling up a concrete ramp shakes like one at 40 km/h, so distance is withheld while climbing and turning together; once that test fired on an ordinary road and dropped 9 seconds at 48 km/h, about 115 m. On 24 journeys the step counter took the engine for footsteps: as recorded, 12.5 km that GPS measured at 10 km/h or more were counted by steps as 1.8 km. On 7 of them that was a fifth of the ride or more, at worst 79%. Stretches counted by steps while GPS showed riding speed are left out of the comparisons (Section 3.5), so this loss is not in the distances reported above.
- **In the air, a typical speed.** The flight engines give the speed an airliner of one type typically has at that point of a flight, not a measurement of this one. A headwind or tailwind, a faster or slower aircraft, a long hold or a short hop will make them read high or low: on NASA's flights the middle 80% counted 91 to 118% of the distance. Examples learned on the phone's own earlier flights with GPS take over when there are some, which one flight could not test. The direction in the air settled about  $12^\circ$  off the track, which carried the drawn end of the flight 16.9 km from GPS's (Figure 7).

## 7.2 What did not work

Table 9 lists approaches that were built and measured against GPS on the same recordings and did no better. Two are worth a note. Leaving the speed out of the angle learner seemed attractive, since it would make direction independent of the speed model, but a car brakes into a corner and accelerates out of it, and a learner driven by the change in turn rate adds the two up instead of cancelling them. And weighting turns by the square root of their push drew individual drives very differently, better on one and worse on another, yet over the 66 rides then

**Table 9.** Approaches that were measured and did not do better, each on the recordings available when it was tried.

| Approach                           | Idea                                                                                                         | Measured against GPS                                                       |
|------------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <i>Direction</i>                   |                                                                                                              |                                                                            |
| Push when speeding up or braking   | In a straight line the push points forward                                                                   | 44° off                                                                    |
| Axis the motorcycle leans about    | The lean axis follows the direction of travel                                                                | 25° off                                                                    |
| Direction of the rider's thigh     | The thigh points forward while seated                                                                        | 73° off                                                                    |
| Angle from turns without the speed | Weight each second by the change in turn rate                                                                | 15–40° off on one drive                                                    |
| Angle from full physics            | Add the speed model's change in speed to the turn                                                            | cars 80% → 48% within 30°                                                  |
| Square-root turn weights           | Let the many small bends count more                                                                          | a tie over 66 rides                                                        |
| Median instead of mean for drift   | Ignore short bursts at stops                                                                                 | walking 93% → 88%                                                          |
| Looser stillness for drift         | Measure drift at more stops                                                                                  | motorcycles 73% → 68%                                                      |
| Magnetic disturbance subtracted    | Earth's field turns with the car, a magnet fixed to the phone does not; fitted with a gyroscope-only heading | leftover 40–200 $\mu$ T: the field changed even on straight roads          |
| Field matched to Earth's numbers   | A second is clean if its strength and dip match Earth's here                                                 | right only ~60% of the time; 3 $\mu$ T or 3° off moved one drive 96% → 68% |
| Core Motion's pulls undone         | In a disturbed field, turn back whatever north the magnetometer adds                                         | no change: after its first seconds it pulls under 5° a minute              |
| <i>Walking, riding and the air</i> |                                                                                                              |                                                                            |
| Thigh swing for walking direction  | Swing is sharper forwards than backwards                                                                     | wrong more often than right                                                |
| Other riding-or-walking tests      | A steady push; speed overruling steps                                                                        | blocked 5% of real walking                                                 |
| Speed from turns in the air        | Sideways push over turn rate                                                                                 | phone's turning hid the aircraft's                                         |
| Integrating the whole climb        | Carry the takeoff integration on                                                                             | worked for one setting only                                                |
| Cabin shaking as a fifth input     | Shaking changes with height and speed                                                                        | 68% → 69% of NASA flights within 10%                                       |
| <i>Speed</i>                       |                                                                                                              |                                                                            |
| Last ten seconds into the network  | More context for each answer                                                                                 | better per second, not per journey                                         |

recorded the learned angle was the same (9.5° against 9.2° median error): single drives are a poor guide to a method.

### 7.3 Lessons

- **The phone's own definitions caused more error than the models.** Its built-in heading becomes unreliable in a common pocket position, drifts in a car and can start wrong; its “user acceleration” points the opposite way to the vehicle's acceleration. Each of these, once found, mattered more than any change to the estimators.
- **GPS can hide in decisions as well as in distance.** The no-GPS mode never drew a metre from GPS, yet at times it used GPS to decide between walking and riding, to set

the first heading, or to carry an angle learned before the mode was switched on. Only reading every place the mode touched GPS found them.

- **Two errors can hide each other.** On one drive a worse angle estimate drew a better route, because it cancelled an error in the heading reference. Judge a part of the method against physics, not only by the route it draws.
- **A replay must do everything the app does,** stops and a phone in the hand included; with those rules in, it differs from what the app drew by a median 7%.
- **A replay must feed each part what the app feeds it.** The flight's direction was at first replayed with the cabin's near-zero vibration speed rather than the speed the app has in the air. That starved the angle learner, and the result in the air swung between  $4^\circ$  and  $24^\circ$  with nothing but the replay's random choices; with the app's own speed it holds at  $10\text{--}13^\circ$ .
- **A replay's own randomness can move results.** When a recording was added, drawing every journey's store afresh moved numbers that had nothing to do with it by several points. Each example now keeps its own fixed chance of being drawn, so a new recording changes each store only by its own examples — which, small as they are, still move single journeys (Section 8).

## 8 Limitations

All recordings come from one person and one iPhone, mostly in one city, with a single flight; other people, phones and vehicles may behave differently, the starting network most of all. The flight engines learned from one aircraft of one regional jet type and were checked on the phone on a single flight of another; the phone measures tilt from gravity, which a steady push can bend, while the recorder measured the aircraft's attitude. Most results replay the newest method on earlier recordings rather than coming from live use, with a speed store drawn from the same person's other journeys; on a short drive the replayed and the live speeds can teach different angles (Section 7.1). Single journeys depend on the store's exact contents: adding one eight-minute car drive changed about 1% of each journey's store and moved single journeys' distance by a median 3.6 points (up to 14), and the motorcycle and car totals by 0.3 and 1.3 points, so the shares of journeys within 20% move by several journeys from one re-run to the next. GPS in a pocket is accurate only to tens of metres, so small errors cannot be measured reliably. In every journey above the phone rode in a pocket, lay flat or sat on a mount. On one car drive it was held in the hand the whole way; recorded live with the current version and kept out of every result, that drive counted 77% of the distance GPS measured and missed the speed by 13.7 km/h, and above 50 km/h it read 31 km/h where GPS said 65. The rule for a phone in the hand applies only while it is being moved: the stretches it marked read 83% of GPS, and those held steadily, which it did not mark, 66%. The magnetic field was recorded on 19 car drives and 12 motorcycle rides so far, and everything said here about it rests on those 32. On the first drive a MagSafe charger sat near the phone for part of the time and the correction worked: 97% of seconds within  $30^\circ$  with the steady-field test, 98% with the turn check. On the second it sat there throughout; the field was steady at an Earth-like strength for only 51 seconds, short of the minute the steady-field test needs, while turning minutes passed the turn check at 27 seconds ( $0\% \rightarrow 43\%$ ). On the third, with the phone placed away from the charger, the field swung between 32 and 74  $\mu\text{T}$  from minute to minute and was never steady; the turn check passed at 7 seconds and the share within  $30^\circ$  moved from 78% to 75%. On the fourth the field rose with speed, from about 70 to 170  $\mu\text{T}$ , and changed even on straight roads, which is why subtracting it as a fixed disturbance did not work (Table 9); no minute passed the turn check, so the steady-field test decides that drive (94%). On the fifth, 52 minutes long, the turn check

passed at 169 seconds, most of them in the first minutes, and put north close to where GPS says it was; but once those samples were twenty minutes old and dropped, a few scattered later ones moved the correction the wrong way for the rest of the drive. The share within  $30^\circ$  stayed high (99%; 100% with no magnetometer at all), because on that drive the angle learned from turns carried an error of the opposite sign. On the sixth the turn check passed at 152 seconds, most of them in the first minutes, and its correction agreed with GPS to within a few degrees, though the drive's share within  $30^\circ$  was 83% against 97% with no magnetometer at all. On the first motorcycle ride the field was steady for 88 seconds and the correction took the graded seconds within  $30^\circ$  from 94% to 97% (96% once the turn check, passing at 5 seconds, takes over), though weak GPS left only 69 seconds to grade; on the second the field stayed near  $500 \mu\text{T}$ , ten times Earth's, nothing could correct the heading, and 28% of its seconds were within  $30^\circ$ . On the third it swung between about 45 and  $150 \mu\text{T}$  while riding and passed the turn check only on the walk afterwards, so the ride's direction rested on its turns alone (81% within  $30^\circ$ ). On the fourth it ran from about 40 to over  $1,000 \mu\text{T}$ , was steady for only 13 seconds and never passed the turn check, and its direction too came from its turns (90%).

## 9 Conclusion

A phone's own sensors can draw a usable route without satellites. Direction is the stronger half: learned from the vehicle's turns on a heading kept honest by its stops and, when possible, by the magnetometer, it stays within  $30^\circ$  of GPS most of the time on every kind of travel measured. Speed from vibration is close in a car and on foot but compressed on a motorcycle. In the air, the takeoff measured by the phone and then a network trained on public airline flight data counted the flight's distance to within 3%, where the takeoff alone counted 61%. The largest remaining error is a constant turn over a whole drive, from where the heading started and from the angle learned for the phone, which so far only a clean magnetic field has been able to reduce.

**Code and data.** The app, including the method described here, is open source at <https://github.com/yu314-coder/GPS-location-app>. The recordings contain the author's locations and are not published; the airline flight data is public [9], and the two flight engines ship with the app.

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## A Every journey

Each motorcycle (M) and car (C) journey, numbered in the order recorded, with no dates. Minutes and GPS km are what GPS could check; app/GPS is the distance ratio for journeys of at least 1 km; speed error is the average over half-minutes; direction is graded where at least 50 seconds could be; the route columns are the whole-route comparison. A dash means too little GPS to grade.

**Table 10.** Every motorcycle (M) and car (C) journey, in the order recorded.

|     | Min. | GPS km | App/GPS<br>(%) | Speed err.<br>(km/h) | Dir. within<br>30° (%) | Dir. median<br>(°) | Route<br>km | Turned<br>(°) | Size |
|-----|------|--------|----------------|----------------------|------------------------|--------------------|-------------|---------------|------|
| M01 | 20   | 5.8    | 64             | 13.3                 | 83                     | 10                 | 13.0        | -2            | 0.60 |
| M02 | 12   | 5.0    | 38             | 25.4                 | 87                     | 8                  | 6.6         | +6            | 0.41 |
| M03 | 20   | 3.5    | 79             | 8.2                  | 48                     | 32                 | 9.9         | -12           | 0.51 |
| M04 | 17   | 1.4    | 34             | 6.7                  | 63                     | 11                 | 5.1         | +5            | 0.74 |
| M05 | 16   | 1.3    | 158            | 6.9                  | 76                     | 19                 | 4.8         | +1            | 0.57 |
| M06 | 19   | 4.1    | 104            | 8.5                  | 44                     | 34                 | 12.3        | +13           | 0.77 |
| M07 | 20   | 5.8    | 72             | 11.4                 | 86                     | 11                 | 8.5         | +1            | 0.67 |
| M08 | 24   | 7.4    | 72             | 10.2                 | 61                     | 26                 | 13.8        | -25           | 0.95 |
| M09 | 9    | 2.0    | 103            | 3.9                  | —                      | —                  | —           | —             | —    |
| M10 | 27   | 9.2    | 82             | 10.1                 | 89                     | 13                 | 14.7        | -14           | 0.84 |
| M11 | 20   | 4.0    | 82             | 7.9                  | 68                     | 9                  | 11.9        | +0            | 0.57 |
| M12 | 14   | 4.8    | 99             | 6.6                  | 83                     | 22                 | 7.3         | +14           | 0.88 |
| M13 | 23   | 6.7    | 96             | 7.7                  | 59                     | 25                 | 11.9        | +24           | 0.90 |
| M14 | 18   | 5.7    | 97             | 10.3                 | 83                     | 12                 | 9.5         | -9            | 0.84 |
| M15 | 18   | 8.2    | 82             | 12.1                 | 89                     | 7                  | 12.0        | -4            | 0.95 |
| M16 | 16   | 5.8    | 84             | 8.6                  | 88                     | 22                 | 8.0         | -25           | 0.80 |
| M17 | —    | —      | —              | —                    | —                      | —                  | 2.5         | -48           | 0.51 |
| M18 | 4    | 1.4    | 99             | 4.6                  | 99                     | 3                  | 2.0         | -3            | 0.84 |
| M19 | 14   | 6.1    | 79             | 8.2                  | 39                     | 33                 | 7.3         | +32           | 0.77 |
| M20 | 19   | 6.2    | 99             | 8.8                  | 90                     | 5                  | 9.3         | -15           | 0.77 |
| M21 | 15   | 3.6    | 100            | 8.1                  | 77                     | 17                 | 8.8         | +9            | 0.71 |
| M22 | 10   | 2.4    | 107            | 7.3                  | 85                     | 8                  | 5.7         | +5            | 0.79 |
| M23 | 11   | 2.3    | 86             | 7.6                  | 80                     | 9                  | 6.5         | -4            | 0.79 |
| M24 | 20   | 3.5    | 87             | 9.9                  | 87                     | 6                  | 12.0        | -6            | 0.93 |
| M25 | 9    | 3.5    | 96             | 6.6                  | 97                     | 4                  | 4.1         | -2            | 1.05 |
| M26 | 27   | 9.7    | 82             | 9.8                  | 76                     | 11                 | 16.6        | -2            | 0.79 |
| M27 | —    | —      | —              | —                    | —                      | —                  | 2.7         | -101          | 0.41 |
| M28 | 14   | 2.0    | 119            | 8.0                  | 62                     | 19                 | 8.0         | -2            | 1.82 |
| M29 | 8    | 2.8    | 79             | 10.0                 | 93                     | 7                  | 4.0         | -8            | 0.71 |
| M30 | 9    | 0.3    | —              | 1.0                  | 26                     | 52                 | 3.2         | -28           | 0.44 |
| M31 | 25   | 7.9    | 64             | 11.9                 | 64                     | 17                 | 15.6        | -15           | 0.70 |
| M32 | 11   | 3.6    | 87             | 8.0                  | 77                     | 22                 | 4.3         | -20           | 0.77 |
| M33 | —    | —      | —              | —                    | 84                     | 13                 | —           | —             | —    |
| M34 | 15   | 1.4    | 109            | 6.9                  | 96                     | 10                 | 5.1         | -7            | 0.52 |
| M35 | 32   | 6.8    | 92             | 6.1                  | 23                     | 33                 | 11.7        | -37           | 0.87 |
| M36 | 12   | 2.6    | 99             | 5.7                  | 65                     | 18                 | 6.0         | -7            | 1.20 |
| M37 | 19   | 6.3    | 95             | 10.7                 | 87                     | 11                 | 13.9        | +10           | 0.75 |

|     | Min. | GPS km | App/GPS<br>(%) | Speed err.<br>(km/h) | Dir. within<br>30° (%) | Dir. median<br>(°) | Route<br>km | Turned<br>(°) | Size |
|-----|------|--------|----------------|----------------------|------------------------|--------------------|-------------|---------------|------|
| M38 | 21   | 4.1    | 95             | 4.7                  | 77                     | 16                 | 11.5        | -23           | 0.77 |
| M39 | 8    | 3.1    | 91             | 8.2                  | 73                     | 24                 | 5.7         | -14           | 1.03 |
| M40 | —    | —      | —              | —                    | 20                     | 53                 | 1.7         | +63           | 1.02 |
| M41 | 27   | 5.7    | 78             | 9.9                  | 57                     | 22                 | 13.1        | -4            | 0.83 |
| M42 | 20   | 4.9    | 99             | 13.7                 | 87                     | 18                 | 10.4        | -15           | 1.04 |
| M43 | 20   | 5.6    | 84             | 9.8                  | 80                     | 10                 | 10.3        | -4            | 0.85 |
| M44 | 7    | 1.2    | 98             | 4.7                  | 96                     | 9                  | 3.1         | -3            | 0.84 |
| M45 | 7    | 1.5    | 123            | 9.5                  | 28                     | 42                 | 3.3         | +17           | 1.09 |
| M46 | 19   | 7.6    | 76             | 11.1                 | 81                     | 6                  | 11.0        | -4            | 0.81 |
| M47 | 17   | 5.5    | 95             | 8.0                  | 90                     | 4                  | 7.5         | +1            | 0.80 |
| M48 | 28   | 5.1    | 112            | 7.2                  | 70                     | 26                 | 13.3        | -20           | 1.15 |
| M49 | 8    | 1.8    | 53             | 18.5                 | 60                     | 10                 | 5.0         | -176          | 1.66 |
| M50 | 20   | 3.7    | 98             | 7.9                  | 84                     | 9                  | 7.9         | +1            | 0.51 |
| M51 | 18   | 5.8    | 94             | 4.9                  | 99                     | 4                  | 7.4         | -3            | 0.85 |
| M52 | 20   | 4.7    | 110            | 7.8                  | 82                     | 12                 | 8.5         | +5            | 0.65 |
| M53 | 12   | 3.0    | 100            | 7.8                  | 70                     | 17                 | 3.8         | -27           | 0.96 |
| M54 | 9    | 1.9    | 110            | 15.0                 | 95                     | 8                  | 3.1         | +15           | 0.96 |
| M55 | 19   | 4.4    | 97             | 7.0                  | 90                     | 12                 | 8.5         | +1            | 0.58 |
| C01 | 11   | 6.6    | 119            | 11.6                 | —                      | —                  | —           | —             | —    |
| C02 | 16   | 7.6    | 136            | 12.9                 | —                      | —                  | —           | —             | —    |
| C03 | 14   | 8.1    | 138            | 21.7                 | —                      | —                  | —           | —             | —    |
| C04 | 9    | 2.7    | 89             | 8.6                  | —                      | —                  | —           | —             | —    |
| C05 | 13   | 6.5    | 62             | 16.1                 | —                      | —                  | —           | —             | —    |
| C06 | 24   | 7.5    | 89             | 8.6                  | —                      | —                  | —           | —             | —    |
| C07 | 16   | 6.5    | 88             | 9.6                  | —                      | —                  | —           | —             | —    |
| C08 | 29   | 6.0    | 118            | 9.1                  | —                      | —                  | —           | —             | —    |
| C09 | 25   | 9.5    | 82             | 7.4                  | 64                     | 17                 | 13.7        | +28           | 0.68 |
| C10 | 24   | 7.4    | 68             | 9.5                  | 89                     | 15                 | 10.9        | +16           | 1.03 |
| C11 | 3    | 0.8    | —              | 3.0                  | 100                    | 5                  | 1.6         | +8            | 1.13 |
| C12 | 21   | 6.5    | 120            | 7.1                  | 97                     | 5                  | 9.0         | -3            | 1.15 |
| C13 | 12   | 3.5    | 124            | 5.9                  | 98                     | 6                  | 4.4         | -6            | 1.20 |
| C14 | 10   | 2.5    | 93             | 2.4                  | 98                     | 20                 | 3.2         | +30           | 0.88 |
| C15 | 12   | 2.6    | 105            | 3.1                  | 70                     | 25                 | 3.1         | -27           | 0.93 |
| C16 | 10   | 1.3    | 97             | 6.2                  | 49                     | 34                 | 2.1         | -14           | 0.57 |
| C17 | 6    | 1.1    | 85             | 9.9                  | 14                     | 46                 | 1.6         | +33           | 0.93 |
| C18 | 7    | 0.9    | —              | 4.1                  | —                      | —                  | 2.0         | +46           | 1.15 |
| C19 | 8    | 1.1    | 133            | 4.5                  | 85                     | 26                 | 2.7         | +33           | 0.90 |
| C20 | 19   | 7.8    | 109            | 6.9                  | 95                     | 15                 | 9.5         | -14           | 1.00 |
| C21 | 6    | 1.2    | 124            | 6.0                  | 78                     | 15                 | 2.5         | +10           | 0.92 |
| C22 | 15   | 9.1    | 95             | 5.4                  | 98                     | 20                 | 10.4        | -14           | 1.03 |
| C23 | 25   | 13.6   | 101            | 5.7                  | 86                     | 25                 | 16.3        | -23           | 0.99 |
| C24 | 8    | 1.1    | 157            | 8.7                  | 97                     | 8                  | 1.8         | +19           | 1.07 |
| C25 | 4    | 0.9    | —              | 7.6                  | 67                     | 10                 | 1.2         | +48           | 0.95 |
| C26 | 15   | 3.0    | 92             | 4.7                  | 70                     | 25                 | 4.7         | +27           | 1.01 |
| C27 | 12   | 4.3    | 120            | 6.5                  | 99                     | 8                  | 4.9         | +11           | 1.18 |
| C28 | 11   | 1.9    | 109            | 3.6                  | 99                     | 12                 | 2.9         | -10           | 1.09 |
| C29 | 18   | 5.5    | 101            | 3.1                  | 1                      | 63                 | 6.5         | +63           | 1.01 |
| C30 | 10   | 2.5    | 98             | 2.4                  | 85                     | 9                  | 2.8         | -10           | 0.97 |
| C31 | 15   | 2.2    | 101            | 2.5                  | 96                     | 10                 | 2.8         | +10           | 1.02 |
| C32 | 11   | 0.0    | —              | 1.2                  | —                      | —                  | —           | —             | —    |
| C33 | 13   | 2.0    | 112            | 3.0                  | 100                    | 6                  | 2.7         | -3            | 1.07 |
| C34 | 32   | 9.2    | 99             | 3.9                  | 99                     | 6                  | 11.0        | -6            | 0.94 |
| C35 | 34   | 10.0   | 101            | 2.7                  | 98                     | 12                 | 11.2        | -13           | 1.00 |
| C36 | 10   | 1.3    | 103            | 2.9                  | 43                     | 33                 | 1.7         | -31           | 0.89 |
| C37 | 8    | 1.7    | 95             | 2.7                  | 75                     | 17                 | 2.1         | -12           | 0.81 |

|     | Min. | GPS km | App/GPS<br>(%) | Speed err.<br>(km/h) | Dir. within<br>30° (%) | Dir. median<br>(°) | Route<br>km | Turned<br>(°) | Size |
|-----|------|--------|----------------|----------------------|------------------------|--------------------|-------------|---------------|------|
| C38 | 49   | 21.9   | 105            | 5.5                  | 94                     | 20                 | 24.5        | -14           | 1.19 |
| C39 | 52   | 22.9   | 96             | 4.6                  | 99                     | 8                  | 25.4        | +2            | 0.83 |
| C40 | 22   | 6.8    | 96             | 6.1                  | 83                     | 6                  | 7.8         | +2            | 0.93 |
| C41 | 35   | 6.9    | 88             | 3.7                  | 95                     | 2                  | 7.9         | -1            | 0.80 |
| C42 | 18   | 5.9    | 114            | 5.5                  | 82                     | 12                 | 6.7         | -23           | 1.21 |
| C43 | 7    | 0.7    | —              | 12.5                 | 100                    | 3                  | 1.4         | -7            | 0.68 |
| C44 | 34   | 7.6    | 93             | 2.9                  | 100                    | 13                 | 8.9         | +13           | 0.91 |
| C45 | 26   | 14.5   | 101            | 3.6                  | 95                     | 16                 | 15.7        | -14           | 0.93 |
| C46 | 11   | 3.4    | 108            | 4.1                  | 91                     | 12                 | 4.0         | +9            | 0.92 |
| C47 | 37   | 12.3   | 100            | 4.9                  | 79                     | 16                 | 15.3        | -22           | 0.93 |
| C48 | 4    | 0.7    | —              | 4.9                  | 96                     | 13                 | 1.2         | -8            | 1.08 |
| C49 | 71   | 41.8   | 68             | 19.2                 | 80                     | 12                 | 50.6        | -30           | 0.49 |
| C50 | —    | —      | —              | —                    | 88                     | 11                 | —           | —             | —    |
| C51 | 16   | 4.5    | 97             | 3.3                  | 90                     | 4                  | 4.9         | -1            | 0.91 |
| C52 | 15   | 4.9    | 100            | 3.0                  | 98                     | 6                  | 5.7         | +5            | 1.00 |
| C53 | 14   | 4.6    | 109            | 3.9                  | 100                    | 9                  | 5.0         | -5            | 1.08 |

## B Settings

**Table 11.** Every setting of the method as evaluated.

| Setting                              | Value                                                                                                                              |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <i>Speed</i>                         |                                                                                                                                    |
| Vibration window for one fingerprint | 5 s at 50 Hz, 11 numbers                                                                                                           |
| Store of the phone's own examples    | up to 4,000                                                                                                                        |
| Neighbours averaged                  | 12                                                                                                                                 |
| Built-in network                     | 11 $\rightarrow$ 32 $\rightarrow$ 32 $\rightarrow$ 1; used until the store holds 3,000, and after that whenever the store declines |
| Network refuses a fingerprint        | farther from all training clusters than 99.5% of training data                                                                     |
| Takeoff integration in the air       | from the last still moment, up to 120 s                                                                                            |
| Flight engines, after the takeoff    | time since the roll; tilt over 60 s, 300 s and since the roll                                                                      |
| Flight network                       | 4 $\rightarrow$ 32 $\rightarrow$ 32 $\rightarrow$ 1, trained on 302 NASA flights                                                   |
| Flight store                         | 4,000 NASA examples, 12 neighbours, weights $1/d^2$                                                                                |
| Walking: a step's echo refused       | while the last 4 s show a clear rhythm (autocorrelation $\geq 0.3$ at 0.3–0.8 s), a peak inside 0.65 of its step period            |
| <i>Direction</i>                     |                                                                                                                                    |
| Riding begins                        | 9 km/h (2.5 m/s)                                                                                                                   |
| Learning the phone's angle           | above 14 km/h (4 m/s); turn explains $\geq 10\%$ of the push                                                                       |
| First estimate                       | after 30 s and a summed speed $\times$ turn rate of 20 m/s <sup>2</sup>                                                            |
| Redrawing a ride                     | when the angle moves $\geq 3^\circ$ , at most every 10 s                                                                           |
| A new ride                           | after 20 s without riding                                                                                                          |
| A phone taken out of the pocket      | angle forgotten after 10 s in the hand                                                                                             |
| Drift measured at stops              | stopped, turning $< 0.005$ rad/s, hand-rotation $< 0.2$ rad/s                                                                      |
| Drift estimate                       | after 30 still seconds, 10-minute memory, at most $3^\circ/\text{min}$                                                             |
| Drift counted only in a field of     | 30–60 $\mu\text{T}$                                                                                                                |
| Clean magnetic field                 | 25–65 $\mu\text{T}$ , strength and dip steady to 3 over 60 s                                                                       |
| North checked by turning             | last 60 s: heading swept $\geq 30^\circ$ , field in the world frame within 8 $\mu\text{T}$ (RMS), 25–70 $\mu\text{T}$              |
| Magnetometer correction              | any turn-checked minute in the last 20 min; before the first, 60 clean seconds; spread $< 15^\circ$                                |
| Walking direction window             | 4 s                                                                                                                                |